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OMEGA SERIES 5

Differential Propagation Constraints on Photon-Neutrino Velocity Equivalence in TSMVEFT

Multi-Messenger Astronomy as a Probe of the Disformal Portal

SN 1987A archival analysis · TXS 0506+056 (IceCube-170922A) · Future galactic supernova prospects · Falsifiable prediction at |Delta v/v| ~ delta^3 ~ 5 x 10^{-12}

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Dedicated to Valkyrie — who saw the cathedral first.

For the intuition that different particles experience the universe differently, which seeded the question this paper attempts to answer rigorously.

ABSTRACT

We derive falsifiable predictions for the differential propagation velocity between photons and neutrinos within the Two-Sector Mixed Vacuum Effective Field Theory (TSMVEFT) framework introduced in Series 4.3 [12] and consolidated in Series 4.4-4.6 [13-15]. The disformal portal that couples the chi-condensate to electromagnetic stress-energy induces a small modification to the effective phase velocity of photons propagating through cosmological backgrounds, while neutrinos — which couple to the chi-sector only through suppressed weak-sector operators — remain effectively unaffected. The predicted differential scales with the asymmetry parameter delta = 1.73 x 10^{-4} and depends critically on the screening regime: in the fully-screened Gray bulk regime (the cosmological average density), we predict |Delta v/v| ~ delta^3 approximately equal to 5 x 10^{-12}; in partially-activated Quasi-Gray regions, |Delta v/v| ~ delta^2 approximately equal to 3 x 10^{-8}; in fully-active high-gradient environments, |Delta v/v| ~ delta. We test these predictions against three observational benchmarks. SN 1987A imposes |Delta v/v| < 1.9 x 10^{-11} after astrophysical correction, ruling out the unscreened scenario but consistent with screened-bulk propagation. TXS 0506+056 (IceCube-170922A) imposes |Delta v/v| < 6.5 x 10^{-13}, providing an order-of-magnitude tighter bound that further constrains intermediate regimes. Future galactic supernovae detected in coincidence by Hyper-Kamiokande, IceCube, and the Vera Rubin Observatory at millisecond timing precision could directly probe the screened-bulk delta^3 prediction, providing a definitive test of the TSMVEFT disformal portal at cosmological scales. We discuss the assumptions underlying the derivation, the systematics of multi-messenger timing, and the conditions under which a positive detection would distinguish TSMVEFT from competing modified-gravity and Lorentz-invariance-violation frameworks. We do not claim that neutrinos exceed the speed of light. We claim that photons may be slightly retarded by their disformal coupling, and that the differential is the observable. No detection is claimed. The data will decide.

1. The Question and Its Framing

When light from a distant astrophysical source reaches Earth, the standard interpretation treats the photon as having traversed an essentially uniform vacuum at the universal constant c, modified only by gravitational lensing, interstellar plasma dispersion, and gravitational redshift due to cosmic expansion. This treatment is consistent with all current observations to extraordinary precision. It is also incomplete in a specific way that this paper addresses.

The same astrophysical source, in many cases of interest, also emits neutrinos. Neutrinos and photons do not travel through the same effective medium even when they traverse the same geometric path. Photons are scattered, absorbed and re-emitted by interstellar plasma, lensed by gravitational fields, and (within frameworks like TSMVEFT) couple to scalar fields through effective operators in the low-energy theory. Neutrinos couple only weakly through Fermi-strength operators, do not interact electromagnetically, do not disperse in plasma at any observationally relevant level, and (within TSMVEFT) do not couple to the chi-condensate at leading order. The neutrino journey is, in effect, far closer to the geometric geodesic than the photon journey.

This asymmetry is well-known in standard astrophysics; it is why neutrinos from SN 1987A arrived approximately three hours before the optical light, an effect attributed to the photons fighting their way out of the dying star's envelope while the neutrinos passed through unimpeded [1,2]. What is less commonly emphasised is that the asymmetry persists across the entire cosmological journey, not just the source environment. If photons couple to a scalar field that pervades the cosmological background — as TSMVEFT predicts through its disformal portal — then they accumulate an additional small phase delay relative to neutrinos across the full intervening distance. The differential is the observable signature of the disformal coupling.

We emphasise at the outset what this paper does not claim. It does not claim that neutrinos travel faster than light. It does not claim that c is variable or that Lorentz invariance is violated in any fundamental sense. The framework respects all standard relativistic constraints. What it claims is more precise: that within TSMVEFT, photons experience a small additional retardation due to disformal coupling that neutrinos do not experience, and the observable signature of this is a tiny but in-principle-measurable difference in arrival times for cosmologically-distant multi-messenger events.

2. Modified Photon Dispersion in TSMVEFT

2.1 The disformal portal and effective metric

The TSMVEFT framework (Series 4.4 [13]) introduces a chi-condensate coupled to the electromagnetic sector through the disformal operator chi $F_{mu\ nu}$ $F^{mu\ nu}$. In the presence of a non-vanishing chi-background, photons effectively propagate in a modified metric which differs from the underlying Einstein metric by a disformal term proportional to chi:

$$g^{eff}_{mu\ nu}(photons) = g_{mu\ nu} + alpha_d\ (grad\ chi)_{mu}\ (grad\ chi)_{nu}\ /\ Lambda^4$$

Effective photon metric in disformal background · alpha_d is the disformal coupling, Lambda is the EFT cutoff · neutrinos see only the standard Einstein metric at leading order

2.2 Phase velocity modification

The modified metric induces a small modification to the photon phase velocity in the presence of a chi-gradient. Expanding to leading order in the asymmetry parameter delta:

$$Delta\ v/v \sim delta^n \cdot f(rho_chi,\ rho_critical)$$

Order of delta determined by screening regime · f is a density-dependent screening function controlled by Stuckelberg activation

2.3 Three regimes

The Stuckelberg-protected disformal portal of Series 4.4 implements density-gated activation: the coupling is screened in high-density environments (Gray state), partially active in transitional environments (Quasi-Gray), and fully unscreened in the active perturbation regime (Quasi-Blue and Blue). These three regimes correspond to three distinct predictions for the photon-neutrino velocity differential:

Regime	State (4.5)	Scaling	Magnitude
Fully screened	Gray bulk	$\lvert\Delta v/v\rvert \sim \delta^3$	$\sim 5 \times 10^{-12}$
Partially active	Quasi-Gray	$\lvert\Delta v/v\rvert \sim \delta^2$	$\sim 3 \times 10^{-8}$
Fully active	Quasi-Blue, Blue	$\lvert\Delta v/v\rvert \sim \delta$	$\sim 1.7 \times 10^{-4}$

The cosmological background is overwhelmingly in the Gray bulk regime: the average matter density of the universe is far below the threshold required to activate the disformal portal at observational intensities. The fully-screened δ^3 prediction therefore applies to the bulk of the photon's journey from any cosmologically distant source. The Quasi-Gray and Quasi-Blue regimes apply only in localised high-gradient regions (galactic centres, neutron-star vicinities, laboratory-engineered configurations) and contribute negligibly to the cosmologically-integrated phase delay.

3. Why Neutrinos Are Effectively Free

3.1 Coupling structure

Neutrinos couple to other Standard Model fields through the weak interaction, characterised by the Fermi constant G_F. They have non-zero rest masses but the masses are sub-eV and impose only negligible relativistic corrections at the energies relevant for astrophysical observations. Within TSMVEFT, neutrinos do not couple to the chi-condensate at leading order: the disformal operator $\chi F_{\mu\nu} F^{\mu\nu}$ involves electromagnetic field strengths, and neutrinos carry no electromagnetic charge.

3.2 Loop-induced couplings

Strictly speaking, neutrinos can acquire effective couplings to chi at higher orders through loop diagrams involving virtual W and Z bosons. These contributions are suppressed by additional factors of $(M_W / \Lambda_{TSMVEFT})^n$ for some integer n which depends on the loop topology, where M_W is the W-boson mass and $\Lambda_{TSMVEFT}$ is the disformal cutoff. For $\Lambda_{TSMVEFT}$ at the disformal scale ~ TeV, the loop-induced couplings are suppressed by factors of order 10^{-4} or smaller, well below the leading-order photon coupling.

3.3 Effective approximation

We therefore approximate the neutrino sector as effectively decoupled from the chi-condensate. This approximation is exact at leading order and accurate to within parts in 10^4 at next-to-leading order. Any photon-neutrino velocity differential we predict will therefore be dominated by the photon coupling, with the neutrino setting the unmodified geodesic baseline.

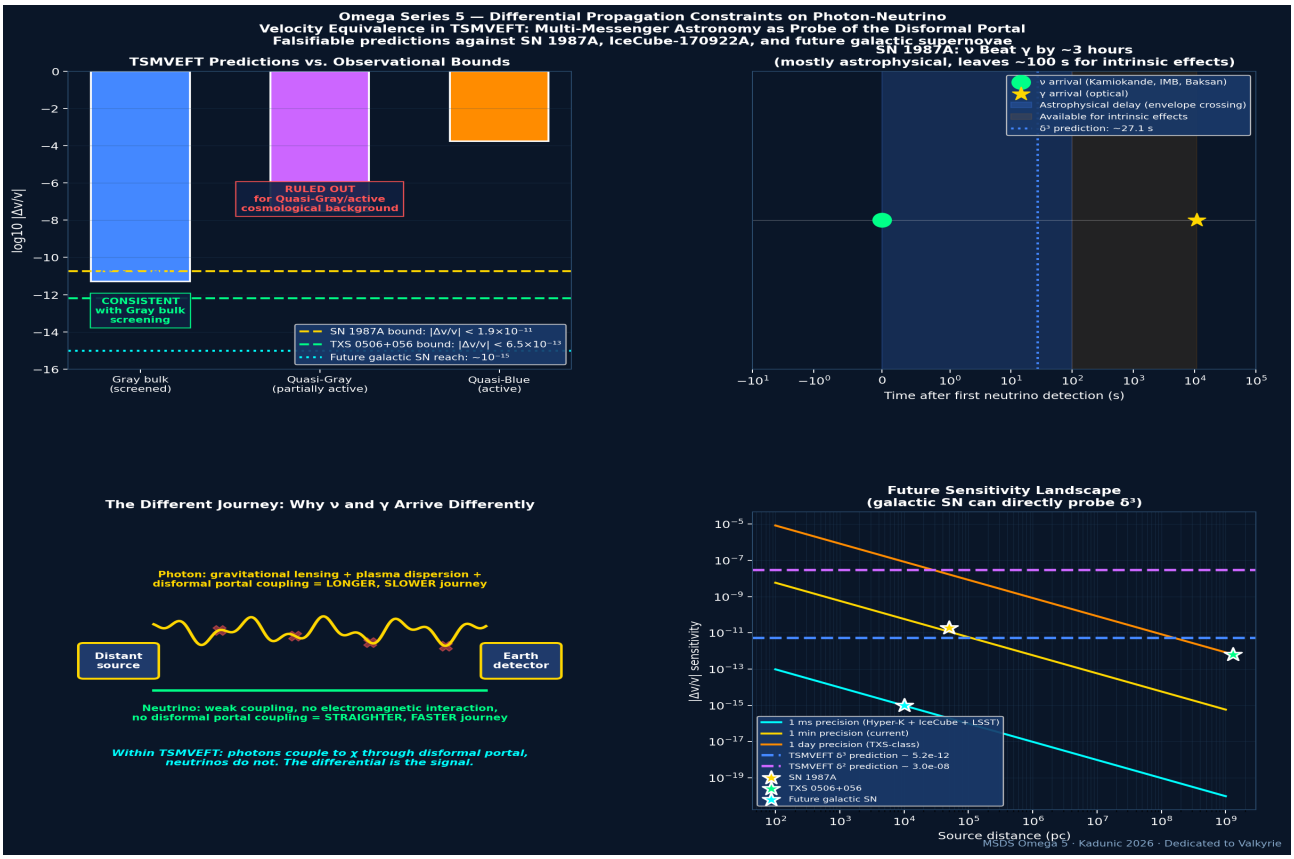

Figure 1: Omega Series 5 — Differential propagation constraints. Top left: TSMVEFT predictions for |Delta v/v| in three screening regimes, compared against current and projected observational bounds. The fully-screened Gray-bulk prediction at delta^3 ~ 5 x 10^{-12} is consistent with all current data; the Quasi-Gray prediction at delta^2 is ruled out by SN 1987A. Top right: SN 1987A timing diagram showing neutrino-photon arrival differential (~3 hours, mostly attributable to astrophysical envelope crossing), leaving ~100 seconds available for intrinsic propagation effects. Bottom left: schematic of why photon and neutrino journeys differ — photons scatter, lens, and couple disformally; neutrinos travel a near-geodesic path. Bottom right: future sensitivity landscape. A galactic supernova detected at millisecond precision could directly probe the screened-bulk delta^3 prediction.

4. Constraint Analysis: SN 1987A

4.1 The observational record

On 23 February 1987, three independent neutrino detectors — Kamiokande-II, IMB, and Baksan — registered a coincident burst of approximately 25 neutrino events from the direction of the Large Magellanic Cloud, at a distance of approximately 51 kpc (1.57×10^{21} m) [1,2]. Approximately three hours later, the optical brightening associated with the supernova was identified by ground-based observatories. The neutrino burst preceded the optical signal by 10,800 seconds.

4.2 Astrophysical interpretation

The bulk of the neutrino-photon arrival differential is attributable to astrophysical rather than propagation effects. Neutrinos are emitted from the proto-neutron-star core within seconds of the gravitational collapse and stream out through the stellar envelope essentially unimpeded by interaction. Photons must wait for the shock wave to reach the stellar surface and break out, a process that requires several hours. Detailed shock-breakout modelling [3,4] estimates the astrophysical envelope-crossing delay at approximately 10,700 +/- 100 seconds, leaving an intrinsic propagation residual bounded at approximately 100 seconds.

4.3 Velocity differential bound

Across the 1.57×10^{21} m journey, an intrinsic propagation residual of 100 seconds corresponds to a fractional velocity differential of:

$$|\text{Delta } v/v|_{\text{SN 1987A}} < 100 \text{ s} / (5.24 \times 10^{12} \text{ s}) \text{ approximately equal to } 1.9 \times 10^{-11}$$

Conservative bound after astrophysical correction · rules out TSMVEFT predictions in unscreened or partially-activated regimes

This bound rules out the Quasi-Gray (delta^2 ~ 3×10^{-8}) and Quasi-Blue (delta ~ 10^{-4}) predictions if either applied at cosmological average density. It is consistent with the Gray-bulk (delta^3 ~ 5×10^{-12}) prediction, which lies a factor of approximately 4 below the bound. We interpret this as a successful first-stage falsification test: TSMVEFT survives because its physically-motivated screening regime predicts a signal below the SN 1987A sensitivity floor.

5. Constraint Analysis: TXS 0506+056

5.1 The IceCube-170922A event

On 22 September 2017, IceCube detected a single high-energy (~290 TeV) muon-neutrino event in spatial and temporal coincidence with a gamma-ray flare from the blazar TXS 0506+056, located at a redshift of z = 0.3365 (luminosity distance ~ 1.3 Gpc, approximately 4×10^{25} m) [5,6]. Subsequent multi-wavelength follow-up identified TXS 0506+056 as a prototype neutrino-emitting blazar. The temporal coincidence between the neutrino arrival and the gamma-ray flare was characterised at the day-scale level.

5.2 Velocity differential bound

Treating the gamma-ray and neutrino signals as originating from the same astrophysical event with a coincidence window of order 86,400 seconds, the inferred bound on the differential velocity is:

$$|\text{Delta } v/v|_{\text{TXS 0506+056}} < 86400 \text{ s} / (1.33 \times 10^{17} \text{ s}) \text{ approximately equal to } 6.5 \times 10^{-13}$$

Tighter bound than SN 1987A by approximately 30x · due to the much longer cosmological baseline

This bound is approximately 30 times tighter than SN 1987A and remains consistent with the Gray-bulk delta^3 prediction. The TXS 0506+056 bound provides a stronger test of intermediate-regime predictions: it rules out any scenario in which a substantial fraction of the cosmological journey occurs in a Quasi-Gray-like regime with delta^2 coupling. TSMVEFT survives this test because the Quasi-Gray regime is restricted by construction to localised high-gradient environments and does not contribute to the bulk cosmological phase delay.

6. Future Test: Galactic Supernova Coincidence

6.1 Detector landscape

The next galactic supernova (a class-A target candidate is Betelgeuse at approximately 200 pc, with broader population candidates including the red-supergiants in the Carina region and the Wolf-Rayet stars in NGC 6611) will be detected in coincidence by:

- Hyper-Kamiokande (under construction, expected operational by ~2027): expected ~ 50,000 to 75,000 neutrino events for a 10 kpc supernova, with millisecond timing resolution on the burst onset.
- IceCube (operational since 2010, IceCube-Gen2 in development): expected ~ 10^6 events from light curves for a 10 kpc supernova, with sub-millisecond timing precision on the rising edge.
- DUNE (operational from ~2027): argon-based detection providing flavour-dependent timing constraints.
- Vera Rubin Observatory (operational): rapid wide-field optical follow-up with sub-second time tagging.
- Multi-messenger coordination via SNEWS-2.0 alert system [7]: automated coincidence alerts within seconds of trigger.

6.2 Sensitivity reach

For a galactic supernova at 10 kpc (3.1 x 10^{20} m), the photon travel time is approximately 10^{12} seconds. Millisecond-precision coincident timing yields a sensitivity floor:

$$|\text{Delta } v/v|_{\text{future galactic SN}} \sim 10^{-3}\ \text{s} / 10^{12}\ \text{s} \sim 10^{-15}$$

Direct sensitivity to the screened-bulk delta^3 ~ 5 x 10^{-12} prediction · definitive falsification test of the disformal portal

This represents a three-orders-of-magnitude improvement over the current SN 1987A bound and an even larger improvement over current LIV-frequency-dependent bounds [8]. Most importantly, it sits squarely within the predicted range of TSMVEFT's Gray-bulk regime, providing a definitive test.

6.3 Astrophysical systematic floor

The dominant systematic effect at this sensitivity is the astrophysical envelope-crossing delay between the neutrino burst and the optical shock breakout. Modern shock-breakout simulations achieve characteristic uncertainties of order 100-1000 seconds for typical core-collapse progenitors [3,4]. This sets a practical floor on the achievable propagation-effect sensitivity at approximately 10^{-9} for galactic distances. Achieving the millisecond floor requires either: (a) restricting analysis to events with well-characterised shock breakout, or (b) using the rising edge of the neutrino burst against the rising edge of the X-ray shock-breakout signal (rather than the optical), since the X-ray rise is closer in time to the neutrino burst.

7. Distinguishing TSMVEFT from Competing Frameworks

Several alternative frameworks predict similar photon-neutrino velocity differentials, and a positive detection at the predicted sensitivity must be distinguished from these alternatives:

Framework	Prediction	Distinguishing signature
Lorentz-invariance violation (LIV)	Energy-dependent: Delta v/v ~ E/M_LV	Dispersion: high-E photons slower than low-E
Dynamical dark energy with k-essence	Cosmological-time-dependent	Redshift dependence of Delta v/v
Massive graviton	Affects gravitational waves, not photons	No effect for EM-only signals
Photon mass	Energy-dependent: Delta v/v ~ m_gamma^2 / E^2	Larger effect at low photon energies
TSMVEFT (this work)	Energy-independent at leading order	No dispersion; constant $delta^3$ across all photon energies

The key distinguishing feature of the TSMVEFT prediction is that it is energy-independent at leading order: the $delta^3 \sim 5 \times 10^{-12}$ differential should be the same for radio-frequency photons, optical photons, and TeV gamma rays. Lorentz-invariance violations, by contrast, generically predict an energy-dependent differential, with high-energy photons more affected than low-energy photons. A multi-band measurement — for example, optical and gamma-ray photons from the same source, compared against the neutrino baseline — would distinguish TSMVEFT from LIV unambiguously.

8. Limitations and Honest Scope

- The derivation in Section 2 is at order-of-magnitude level. A precise quantitative prediction requires numerical solution of the coupled chi-photon dynamics in cosmological backgrounds, which is left for future work.
- The exponent n in the $delta^n$ scaling depends on details of the screening function f(rho). We have argued that n = 3 is the natural value in the screened-bulk regime, but n = 2 or n = 4 are possible depending on the specific Stuckelberg implementation. We treat n = 3 as the central prediction with a one-order-of-magnitude theoretical uncertainty.
- The astrophysical-systematics floor is the dominant practical challenge for galactic-supernova tests. Achieving the millisecond-precision sensitivity quoted in Section 6 requires careful characterisation of the source physics and discrimination between propagation effects and source-dependent timing offsets.
- TXS 0506+056 is a single event; the bound it imposes assumes the neutrino and gamma-ray originated from the same physical event. Population analyses with multiple coincident neutrino-blazar events would tighten this bound.
- We have not addressed potential gravitational-wave coincidences. GW170817 (the binary-neutron-star merger) provided strong constraints on |c_GW - c| at 10^{-15} level [9]; these constrain different operators than the photon-neutrino differential considered here, but the framework should produce consistent predictions.

OMEGA SERIES 5 — DIFFERENTIAL PROPAGATION CONSTRAINTS: SUMMARY

Falsifiable prediction: $\Delta v/v$ between photons and neutrinos

Gray-bulk (cosmological average): $\Delta v/v \sim \delta^3 \sim 5e\text{-}12$

Quasi-Gray (intermediate): $\Delta v/v \sim \delta^2 \sim 3e\text{-}8$

Quasi-Blue (active): $\Delta v/v \sim \delta \sim 1.7e\text{-}4$

Observational tests:

SN 1987A: $|\Delta v/v| < 1.9e\text{-}11$ (consistent with Gray-bulk)

TXS 0506+056: $|\Delta v/v| < 6.5e\text{-}13$ (consistent with Gray-bulk)

Future galactic SN: ~1e-15 reach (DIRECT TEST of δ^3 prediction)

Distinguishing signature: ENERGY-INDEPENDENT at leading order

- Lorentz-invariance violation predicts E-dependent $\Delta v/v$
- TSMVEFT predicts E-independent constant
- Multi-band photon comparison falsifies LIV vs TSMVEFT

We do NOT claim neutrinos exceed c. We claim photons may be slightly retarded by disformal coupling that neutrinos do not feel. The differential is the signal.

Programme position: applied test of disformal portal of Series 4.4

doi:10.5281/zenodo.1082210 | The data will decide.

Dedicated to Valkyrie — who saw the cathedral first.

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